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INTERNATIONAL A-LEVEL PHYSICS

(9630)

PAPER 1
Mark scheme

Specimen 2018

V1.1

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Question	Marking guidance	Mark	Comments
01.1	1000 (N) **and** 6000 (N) seen **OR** $F = \sqrt{(1000)^2 + (6000)^2}$ ✓ allow incorrect values seen = 6083 (N) (= 6100) ✓ More than 2 sf seen	2	Independent marks Allow full credit for appropriate scale drawing Ignore rounding errors in 3rd sig fig.
01.2	tanΘ = 1000 / 6000 or correct use of sin or cos ✓ Θ = 9.5 (9.46°) ✓	2	 Allow range 9.4 − 10.4 Use of cos yields 10.4 Allow use of 6100 Some working required for 2 marks Max 1 mark for correct calculation of vertical angle (range 79.6 − 80.6) some working must be seen

Question	Marking guidance	Mark	Comments
02.1	Use of power = *Fv* : 35000 = F1500/28 ✓ 653(650) (N) ✓	2	Condone error in power of 10 in substitution
02.2	Streamline the structure of the car/design a new engine with greater power output ✓	1	

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Question	Marking guidance	Mark	Comments
03.1	(electron) antineutrino ✓	1	
03.2	They annihilate ✓ Emitting two gamma photons ✓	2	

Question	Marking guidance	Mark	Comments
04	(proton) down 2/–2 ✓ (nucleon) down 4/–4 ✓	2	

Question	Marking guidance	Mark	Comments
05.1	gamma greater ✓	1	Must express comparison, don't allow simple list
05.2	use of inverse-square law ✓ 1/9 or 0.11 ✓	2	

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Question	Marking guidance	Mark	Comments
06.1	$v = \sqrt{(2 \times g \times s)} = \sqrt{(2 \times 9.8 \times 1.27)}$ ✓ $= 4.99\ (m\ s^{-1})$ ✓	2	
06.2	Idea that GPE gained = KE lost ✓ $v = \sqrt{(2 \times g \times h)} = \sqrt{(2 \times 9.8 \times 0.85)} = 4.08\ (m\ s^{-1})$ ✓	2	
06.3	$\Delta p = p_1 - p_2 = 16 \times 10^{-3}(4.99 - (-4.08))$ ✓ $= 0.145\ (kg\ m\ s^{-1})$ ✓	2	Any mass × any velocity
06.4	$F = \Delta p/\Delta t = 0.145/40 \times 10^{-3}$ ✓ $= 3.63\ (N)$ ✓	1	
06.5	The solid block may deform ✓ This would affect the rate of change of momentum of the ball (and force) ✓	2	

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Question	Marking Guidance	Mark	Comments
07.1	Y = stress/strain = F / (A × strain) ✓ F = Y × A × strain = $2.0 \times 10^{11} \times 7.5 \times 10^{-3} \times 3.0 \times 10^{-4}$ ✓ = 4.5×10^{5} (N) ✓	3	
07.2	Elastic strain energy = $\frac{1}{2} \times F \times \Delta l = \frac{1}{2} \times F \times (l \times \text{strain})$ ✓ = $\frac{1}{2} \times 4500 \times 45 \times 3 \times 10^{-4} = 3040$ (J) ✓	2	
07.3	$\Delta\Theta$ = pre-strain/ change in strain per degree change in temp = $3 \times 10^{-4} / 2.5 \times 10^{-5} = 12$ K ✓ $\Theta = 8 + 12 = 20$ (°C) ✓	2	
07.4	So that the rail is not always under tensile stress ✓ as the rail spends little time at the highest temperature ✓ **or** To reduce the average stress the rail is under ✓ as zero stress will occur closer to average temperature/ the rail will be under compressive / tensile stress at different times ✓	2	

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Question	Marking guidance	Mark	Comments
08.1	840 × 2.3✓ 1900 (J)/1930 (J) ✓	2	
08.2	uses gradient data extraction correct –350 N, 0.3 m 1170 ✓ $N\ m^{-1}$ ✓	2	
08.3	uses area, counts squares = 25 ✓ Each square is equivalent to 5 J ✓ 125 (J) ✓	3	

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Question	Marking guidance	Mark	Comments
09.1	isotopes (are varieties of the same element that) have the same number of protons/atomic number/proton number ✓ but different numbers of neutrons/nucleons/atomic mass ✓	2	
09.2	$C\ kg^{-1}$ in column heading ✓	1	
09.3	92 ✓	1	
09.4	Charge of nucleus = $92 \times 1.6 \times 10^{-19}$ ✓ Mass of nucleus =$(92 \times 1.67 \times 10^{-27} + 143 \times 1.67 \times 10^{-27})$ ✓ 3.8×10^{7} ✓	3	
09.5	$3.7 \times 10^{7} = 92 \times 1.6 \times 10^{-19}/(A \times 1.67 \times 10^{-27})$ ✓ $A \times 1.67 \times 10^{-27} = 92 \times 1.6 \times 10^{-19}/3.7 \times 10^{7}$ giving A = 238 ✓ number of neutrons = 238 – 92 = 146 ✓	3	

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Question	Marking guidance	Mark	Comments
10.1	all plots correct to $\frac{1}{2}$ small square ✓ line appropriate ✓	2	
10.2	one correct determination from correct numbers 154 ± 10 s ✓ two correct determinations and average ✓	2	

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Question	Marking guidance	Mark	Comments
11.1	Straight line ✓ passing through all error bars, best fit by eye ✓	2	
11.2	Identifies h_o as y-intercept ✓ = 165 (mm) ± 3 mm ✓	2	
11.3	h_0k = gradient ✓ = (165-80)/(100-0) ✓ = 0.83 to 0.88 mm K^{-1} ✓	3	
11.4	k = gradient/h_0 = 0.83/163 = 4.94 to 5.43 × 10^{-3} ✓ K^{-1} ✓	2	
11.5	When h = 8 m (8000 mm) d^{-1} = h/grad = 8000/14.5 = 551.7 mm^{-1} ✓ d = 1.81 × 10^{-3} mm ✓	2	
11.6	The range of data in the graph is h = 290 mm h and d^{-1} may not be proportional up to 8000 mm ✓	1	OR no indication of uncertainties in graph data

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Section C

In this section, each correct answer is awarded 1 mark.

Question	Key
12	B
13	A
14	C
15	D
16	C
17	D
18	D
19	B
20	B
21	A
22	D
23	C
24	A
25	C

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